

Statistical Analysis Report

1. Introduction

This statistical analysis was conducted on the Heart Disease Prediction dataset to identify significant relationships between patient characteristics and the presence of heart disease. The analysis supports feature selection and provides evidence for variables that may be useful in the development of a machine-learning model. A significance level of 0.05 was used. Therefore, results with a p-value below 0.05 were considered statistically significant.

Four main statistical tests were conducted: Pearson correlation, Chi-square test, independent samples t-test, and one-way ANOVA.

2. Statistical Tests

- **Pearson Correlation: Age and Oldpeak**

Objective: To determine whether Age and Oldpeak have a statistically significant linear relationship.

H_0 : There is no significant linear relationship between Age and Oldpeak.

H_1 : There is a significant linear relationship between Age and Oldpeak.

The p-value was compared with 0.05 to determine whether the relationship was statistically significant. A significant result would indicate that age and exercise-related ST depression are associated within this dataset. This relationship may provide useful information for predictive modelling, although correlation does not imply causation.

- **Chi-Square Test: Chest Pain Type and Heart Disease**

Objective: To determine whether ChestPainType is associated with HeartDisease.

H_0 : Chest pain type and heart disease status are independent.

H_1 : Chest pain type and heart disease status are associated.

If the p-value is below 0.05, the null hypothesis is rejected, indicating a statistically significant association between chest pain type and heart disease. This makes ChestPainType a potentially important feature for the machine-learning model and suggests that different chest pain categories may be associated with different levels of heart disease risk.

- **Independent Samples T-Test: Maximum Heart Rate**

Objective: To determine whether the mean Maximum Heart Rate differs between patients with and without heart disease.

H_0 : The mean MaxHR is the same for both groups.

H_1 : The mean MaxHR differs between the two groups.

A p-value below 0.05 indicates a statistically significant difference in maximum heart rate between the two groups. Such a difference would suggest that MaxHR may provide useful predictive information when distinguishing patients with and without heart disease.

- **One-Way ANOVA: Oldpeak and Chest Pain Type**

Objective: To determine whether the mean Oldpeak differs across ChestPainType categories.

H_0 : Mean Oldpeak is equal across all chest pain categories.

H_1 : At least one chest pain category has a different mean Oldpeak.

If $p < 0.05$, the null hypothesis is rejected, indicating that Oldpeak differs significantly across at least some chest pain categories. This may provide additional evidence that exercise-related cardiac measurements and chest pain characteristics contain useful predictive information.

4. Overall Findings and Business Implications

The statistical analysis provides evidence about relationships between demographic, clinical and exercise-related variables and heart disease. The results will be used together with exploratory data analysis and feature engineering when selecting variables for machine-learning development.

Features that demonstrate statistically significant relationships, strong associations, or meaningful predictive information will be considered for retention. However, statistical significance alone will not determine whether a feature is selected. Correlation between predictors, redundancy, domain relevance and potential predictive value will also be considered.

From a business perspective, identifying important patient characteristics can support the development of data-driven heart disease risk prediction systems. Such systems could potentially help healthcare professionals identify patients who may require further assessment.

It is important to note that these results represent statistical associations rather than causal relationships. The dataset is also limited in size and may not represent all patient populations. Therefore, further validation and machine-learning evaluation are required before drawing conclusions about real-world clinical use.

5. Conclusion

The statistical analysis strengthened the understanding of relationships within the Heart Disease Prediction dataset. Pearson correlation, Chi-square, t-test, and ANOVA provided complementary evidence about numerical and categorical variables.

The findings will support feature selection and the development of the machine-learning models. Variables showing meaningful statistical and predictive relationships will be carried forward, while redundant or less informative variables will be reviewed before final model development.